



Programme: B.E – CSE (AI&ML) and CSE(Cyber Security)

Internal Assessment – II

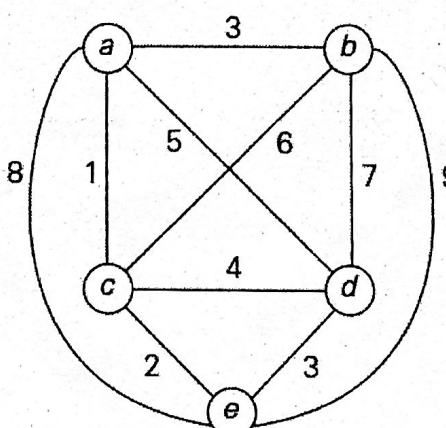
TERM: 15/4/2024 to 27/7/2024	COURSE NAME: Design and Analysis of Algorithms
DATE: 23/07/2024 TIME: 9:30am to 10:30am	COURSE CODE: CI43/CY43
MAX MARKS: 30	PORTIONS: L21-L42
SEMESTER: 4 th SEM	SECTION: A



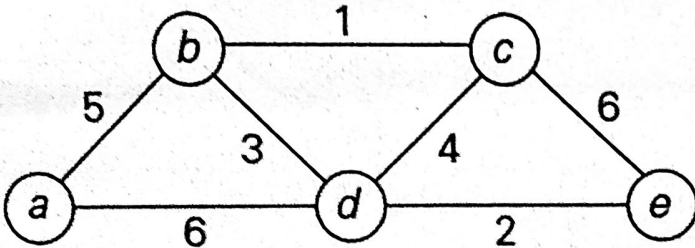
Mobile Phones are banned

Instructions to Candidates: Answer any TWO full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels	CO	Marks																		
1.a	For the set of items given in Table 1a and knapsack capacity = 6 kg, find the optimal solution for the 0/1 knapsack problem making use of dynamic programming approach. Table 1a. Knapsack Table <table><tr><th>item</th><th>weight</th><th>value</th></tr><tr><td>1</td><td>3</td><td>\$25</td></tr><tr><td>2</td><td>2</td><td>\$20</td></tr><tr><td>3</td><td>1</td><td>\$15</td></tr><tr><td>4</td><td>4</td><td>\$40</td></tr><tr><td>5</td><td>5</td><td>\$50</td></tr></table> capacity $W = 6$.	item	weight	value	1	3	\$25	2	2	\$20	3	1	\$15	4	4	\$40	5	5	\$50	L4	CO4	6
item	weight	value																				
1	3	\$25																				
2	2	\$20																				
3	1	\$15																				
4	4	\$40																				
5	5	\$50																				
b	Differentiate between fixed length encoding and variable length encoding in data compression technique. Illustrate the Huffman variable length encoding considering the symbol and frequency given. Characters and Frequencies: a: 20, c:11, d:2, e: 10, o:15, m:8, s:10, t:22, u: 2	L4	CO4	5																		
c	Differentiate NP Hard and NP Complete problems.	L3	CO5	4																		
2.a	Apply dynamic Programming to find the sum of subset of the set of elements given $S = \{1,3,5,7\}$, Sum= 8.	L4	CO4	5																		
b	Demonstrate the traveling salesman problem by constructing the State Space Tree for the graph given in Fig 2b. Consider the source vertex as 'a'.  Fig 2b Graph	L3	CO3	6																		



c	Apply Prim's algorithm to the graph given in Fig 2c. Include in the priority queue all the vertices not already in the tree. Consider source vertex as 'b'.  Fig 2c Graph	L3	CO3	4																								
3.a	Describe the memorization Approach used in Dynamic Programming by illustrating the recursive sum of n natural numbers function. Construct the control stack and illustrate the process of Memorization table. Program: <pre>int main() { int num; printf("Sum = %d", addNumbers(6)); return 0; } int addNumbers(int n) { if (n != 0) return n + addNumbers(n - 1); else return n; }</pre>	L3	CO4	5																								
b	Identify and compute the optimal set of intervals with maximum value for the set of intervals and weights given in Table 3b. Table 3b Weighted Interval <table><tr><th>Interval</th><th>Start time</th><th>Finish time</th><th>Weight</th></tr><tr><td>1</td><td>1</td><td>6</td><td>10</td></tr><tr><td>2</td><td>2</td><td>4</td><td>3</td></tr><tr><td>3</td><td>5</td><td>9</td><td>4</td></tr><tr><td>4</td><td>7</td><td>14</td><td>20</td></tr><tr><td>5</td><td>11</td><td>18</td><td>2</td></tr></table>	Interval	Start time	Finish time	Weight	1	1	6	10	2	2	4	3	3	5	9	4	4	7	14	20	5	11	18	2	L3	CO4	6
Interval	Start time	Finish time	Weight																									
1	1	6	10																									
2	2	4	3																									
3	5	9	4																									
4	7	14	20																									
5	11	18	2																									
c	Solve the given LPP by Simplex Method: Maximize $z= 2x-y+z$ Subject to constraints: $2x+y\leq10$ $X+2y-2z\leq20$ $X+2z\leq5$	L4	CO5	4																								

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create